

Proof.

$$\begin{aligned}
 \text{Ex}[R] &= \sum_{r \in \text{range}(R)} r \cdot \Pr[R = r] && \text{(by 18.3)} \\
 &= \sum_r r \cdot \sum_i \Pr[R = r \mid A_i] \Pr[A_i] && \text{(Law of Total Probability)} \\
 &= \sum_r \sum_i r \cdot \Pr[R = r \mid A_i] \Pr[A_i] && \text{(distribute constant } r) \\
 &= \sum_i \sum_r r \cdot \Pr[R = r \mid A_i] \Pr[A_i] && \text{(exchange order of summation)} \\
 &= \sum_i \Pr[A_i] \sum_r r \cdot \Pr[R = r \mid A_i] && \text{(factor constant } \Pr[A_i]) \\
 &= \sum_i \Pr[A_i] \text{Ex}[R \mid A_i]. && \text{(Def 18.4.4 of cond. expectation)}
 \end{aligned}$$

■

18.4.6 Mean Time to Failure 平均失效时间

A computer program crashes at the end of each hour of use with probability p , if it has not crashed already. What is the expected time until the program crashes? This will be easy to figure out using the Law of Total Expectation, Theorem 18.4.5. Specifically, we want to find $\text{Ex}[C]$ where C is the number of hours until the first crash. We'll do this by conditioning on whether or not the crash occurs in the first hour.

So define A to be the event that the system fails on the first step and $\bar{A}$ to be the complementary event that the system does not fail on the first step. Then the mean time to failure $\text{Ex}[C]$ is

$$\text{Ex}[C] = \text{Ex}[C \mid A] \Pr[A] + \text{Ex}[C \mid \bar{A}] \Pr[\bar{A}]. \quad (18.5)$$

Since A is the condition that the system crashes on the first step, we know that

$$\text{Ex}[C \mid A] = 1. \quad (18.6)$$

Since $\bar{A}$ is the condition that the system does *not* crash on the first step, conditioning on $\bar{A}$ is equivalent to taking a first step without failure and then starting over without conditioning. Hence,

$$\text{Ex}[C \mid \bar{A}] = 1 + \text{Ex}[C]. \quad (18.7)$$